

Fluid Flow Through Fluidized Beds

Pre-Lab Plan
Unit Operations Lab 3 
10/1/2019

Group 1, Tuesday, PM section
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1. Experimental Objective

The objective of this lab is to study the fluidization behavior of a bed of two types of small particles, a mixture of fine silica sand and ion exchange resin beads. The minimal fluidization velocity is determined experimentally by measuring the pressure drop across the bed and then compared to predicted values from using the Ergun equation.

2. Experimental

2.1 Apparatus

The apparatus is the fluidized bed vessel consisting of 4" plexiglass tube with an enlargement at the top, that will be removed during the loading of the bed material. Main school pressurized air supply is used to provide air flow into the system. Next to the manometer, which measures the pressure drop across the bed, are air flow valves used to adjust the air flow.

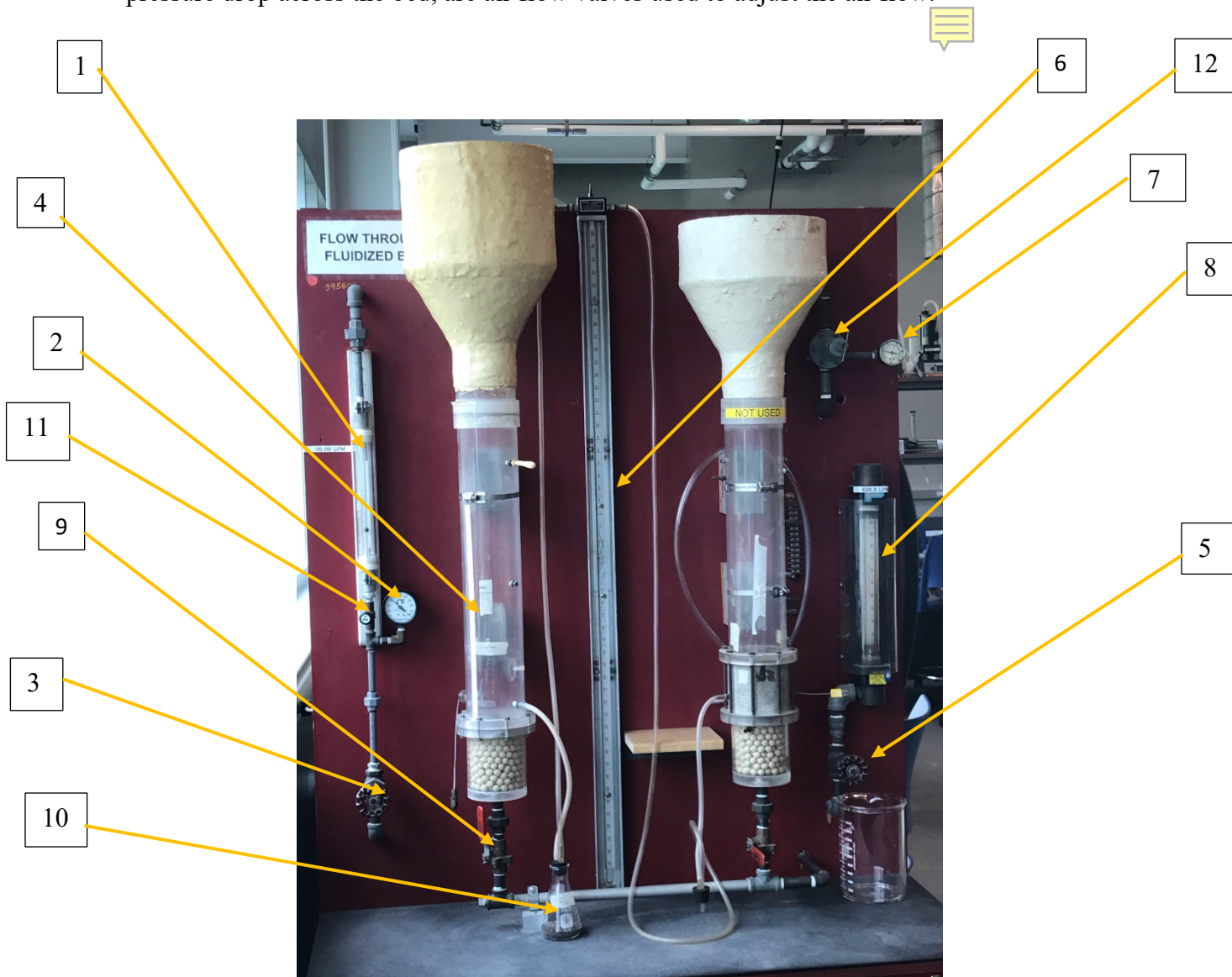


Figure 1: Fluidized bed apparatus



Figure 2: Back side of the fluidized bed apparatus

Component	Function
1 – Rota meter	Measures volumetric flow rate of fluid in the 4" plexiglass tube.
2 – Pressure gauge	Indicates pressure inside the 4" tube.
3 – Air flow control valve	Regulates the flow of air into the tube.
4 – 4" Plexiglass tube (Fluidized bed vessel)	Vessel that contains the bed material, silica sand and the ion exchange bead. Air is introduced into the bottom.
5 – Air flow control valve	Regulates the flow of air into the tube.
6 - Manometer	Measures pressure drop across the bed.
7 – Pressure gauge	Indicates the initial pressure coming into the system from the main air supply.
8 – Rota meter	Measures volumetric flow rate of fluid in a closed tube.
9 – Air inlet valve	Opens and shuts off air flow into the vessel.
10 – Sand trap	Prevents sand from entering the manometer by trapping sand from the bed.
11 – Needle valve	Allows precise regulation of flow.
12 – Air inlet valve	Allows air flow into the system.
13 – Air inlet valve	Allow air flow into the system from the main air supply (120 psig).
14 – Tubing	Connects pressurized air from the main air supply.

Table 1. Description of main components and their functions. Only left side of system is used.



2.2 Materials and Supplies

	Material	Description
1	Silica sand	First material used to charge the bed.
2	MBD 15 ion exchange resin beads	Second material to charge the bed.
3	Air	Gas that flows through the fluidized bed vessel.
4	Set of sieves	Used to determine the particle size distribution of the sand.
5	Safety goggles	Protection for eyes.
6	Nitrile gloves	Protect hands.
7	Water	Used to find void fraction at minimum fluidization
8	Balance	Measures mass for determining density of sand and resin beads as well as for finding particle size distribution of sand and beads.
9	Vacuum	Used to clean the vessel of the bed material. There are 2: one for sand and one for resin beads.
10	Graduated cylinder	Volume measuring device used for determining density and void fraction.
11	Ruler	Measures height of bed material.

Table 2: Materials and supplies with corresponding functions

2.3 Experimental Plan

Fluidized Bed Procedure

- 1.) Become familiarized with the fluidized bed lab setup.
- 2.) Obtain the fine silica sand and the resin beads to conduct the two sets of experiments with the fluidized bed.
- 3.) Start with the silica sand and insert sand up to 6 inches in the bed.
- 4.) Slowly open the needle valve that opens the air to adjust the flow rate and record each pressure drop as a function of gas flowrate.
- 5.) Continue to increase the gas flow until conditions beyond minimum fluidization are reached.
- 6.) One beyond that gas flow, start to decrease the gas velocity and record the pressure drop across the bed as the gas velocity is decreased to zero.
- 7.) Take measurements for the static bed height of 10 inches for the sand now. Then repeat this procedure with the next bed material which is the resin beads. We need one trial for each
- 8.) Plot the measured pressure drop per unit bead height as a function of the gas velocity for both increasing and decreasing flowrates for each bed material.

Sand Particle Size Determination Procedure

- 1.) The sand consists of a distribution of particle size. Thus, start by acquiring the different sieves sizes from the cabinet.
- 2.) Next, measure the mass of a specific amount of sand and put it onto to sieve on the top.
- 3.) Start shaking the sieves to separate and place the different sizes of sand in their respective sieves.

- 4.) Measure and record the amount of sand in each sieve.
- 5.) Using the mass of sand acquired by each sieve, determine the distribution of each particle size and obtain the average particle size.



Void Fraction Determination Procedure

- 1.) Start by filling the graduated cylinder with 50mL of sand.
- 2.) Next, to fill up the interstitial voids, fill graduated cylinder with water until the water is filled up to the 50ml line in the graduated cylinder.
- 3.) Record the volume of water added to the graduated cylinder- this volume will be the void volume.
- 4.) Use the void volume to obtain the void fraction.
- 5.) Repeat this process to obtain the void volume for the resin beads.

Independent and dependent variables

The independent variables are the flow rate of the air through the bed, the bed material, the static bed height of the bed material, and the column size.

The dependent variables are the pressure drop and the minimum fluidization velocity.

The Ergun equation will be used to obtain the pressure drop by using the experimental parameters provided in equation 1.



$$\frac{\Delta P}{L} = 150\bar{V}_0 \frac{\mu}{\Phi_s^2 D_p^2} \left(\frac{(1 - \varepsilon)^2}{\varepsilon^3} \right) + \frac{1.75\rho\bar{V}_0^2}{\Phi_s D_p} \left(\frac{1 - \varepsilon}{\varepsilon^3} \right) \quad \text{Equation 1}$$

For small particles and for laminar flow models- equation 2 can be used to calculate the minimum fluidization velocity. The calculated and experimental values should be close in value.

$$\bar{V}_{0M} \approx \frac{g(\rho_p - \rho)}{150\mu} \frac{\varepsilon_M^3}{(1 - \varepsilon_M)} \Phi_s^2 D_p^2 \quad \text{Equation 2}$$

Project Safety Evaluation

Project Title: Fluid Flow Through Fluidized Beds

Date: 10/1/19

	<i>Yes/No</i>	
Potential electrical hazards?	No	If yes, identify conditions:
Potential mechanical hazards?	Yes	If yes, identify conditions: The process of using a ladder to go up to charge the vessel using bed material can be dangerous. Also, too much pressure can potentially break the manometer.
Potential pressure hazards?	Yes	If yes, identify conditions: Pressure gauges may be over pressured and can potentially break leading to injuries. Other hazards like leakage of the valve and failure of the air regulator can also be unsafe.
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: MBD-15 can cause stomach pains if large amounts digested. It can also cause eye and skin irritation. Silica sand is an eye and inhalation hazard.
Gloves required?	Yes	If yes, specify type: Nitrile gloves
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: MBD-15 CAS#: 69011-18-3 Silica Sand CAS#: 14808-60-7 Compressed Air CAS# 132259-10-0
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	Room Temperature	N/A
Pressure (psia)	40 psi	Pressure entering, pressure going to the flowmeter, and the pressure moving to column
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
	Yes	Pressure gauges, and main shutoff valve for main compressed air entering apparatus.

I have read relevant background material for the Unit Operations Laboratory entitled: "Fluid Flow Through Fluidized Beds" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Marium Murtaza 10/1/19
 signature of team member date

Adisa Hadzic 10/1/19
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Cheryl Slykas 10/1/19
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